Linear Algebra and Its Applications

Section 5.1: Eigenvectors and Eigenvalues

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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5.1 Eigenvectors and eigenvalues

An eigenvector of A is a *nonzero* vector v for which $Av = \lambda v$ for some scalar λ. The scalar is the eigenvalue: the direction stays on the same line, while the vector is scaled, possibly reversed or sent to zero. Rearranging gives

$$(A - \lambda I)v = 0.$$

Thus the eigenspace E_λ is $\mathrm{Nul}(A - \lambda I)$, and λ is an eigenvalue exactly when this null space contains a nonzero vector. The eigenspace includes zero; zero itself is never an eigenvector. To find eigenvectors for a specified eigenvalue, reduce $A - \lambda I$, not A. Row reduction of A itself generally changes its eigenvalues.

Practice problem 5.1.P1: Test five without guessing an eigenvector

The supplied matrix gives

$$A - 5I = \begin{bmatrix} 1 & -3 & 1 \\ 3 & -5 & 5 \\ 2 & 2 & 1 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - 3R_1$, $R_3 \leftarrow R_3 - 2R_1$:

$$\begin{bmatrix} 1 & -3 & 1 \\ 0 & 4 & 2 \\ 0 & 8 & -1 \end{bmatrix}.$$

Then $R_3 \leftarrow R_3 - 2R_2$ gives last row $(0, 0, -5)$. There are three pivots and no free variables, so $(A - 5I)x = 0$ has only zero as a solution. Hence 5 is not an eigenvalue.

Practice problem 5.1.P2: A power acting on an eigenvector

If $Ax = \lambda x$, then $A^2x = A(\lambda x) = \lambda Ax = \lambda^2 x$. Applying A once more gives $A^3x = \lambda^2 Ax = \lambda^3 x$. Each step uses linearity to pull out a scalar. Repeating the argument proves $A^k x = \lambda^k x$ for every nonnegative integer k.

Practice problem 5.1.P3: Several independent vectors within one eigenspace

Yes, all four vectors are independent. Suppose $c_1b_1 + c_2b_2 + c_3b_3 + c_4b_4 = 0$. Apply $A - \lambda_3 I$:

$$c_1(\lambda_1 - \lambda_3)b_1 + c_2(\lambda_2 - \lambda_3)b_2 = 0.$$

The two vectors b_1, b_2 correspond to distinct eigenvalues and are independent by Theorem 2. Since the eigenvalue differences are nonzero, $c_1 = c_2 = 0$. The original relation now reduces to $c_3b_3 + c_4b_4 = 0$, and their assumed independence forces $c_3 = c_4 = 0$. This proof also handles the possibility that $c_3b_3 + c_4b_4$ is zero; zero must not be called an eigenvector.

Exercise 5.1.1: Test a proposed eigenvalue

What we must show. Decide whether $\lambda = 2$ has a nonzero eigenvector. Reduce $A - \lambda I$, shown here as B:

$$B = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (3)R_1$. Then

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}.$$

The reduced equation is $x_1 + 2x_2 = 0$, with x_2 free. Taking $x_2 = 1$ gives $v = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \neq 0$. Directly $Av = \begin{bmatrix} -4 \\ 2 \end{bmatrix} = 2v$.

Conclusion. Yes, 2 is an eigenvalue; all nonzero multiples of v are corresponding eigenvectors.

Exercise 5.1.2: Test a proposed eigenvalue

What we must show. Decide whether $\lambda = -3$ has a nonzero eigenvector. Reduce $A - \lambda I$, shown here as B:

$$B = \begin{bmatrix} 2 & 4 \\ 6 & 12 \end{bmatrix}.$$

Use $R_1 \leftarrow \frac{1}{2}R_1, \quad R_2 \leftarrow R_2 - (6)R_1$. Then

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}.$$

The reduced equation is $x_1 + 2x_2 = 0$, with x_2 free. Taking $x_2 = 1$ gives $v = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \neq 0$. Directly $Av = \begin{bmatrix} 6 \\ -3 \end{bmatrix} = -3v$.

Conclusion. Yes, -3 is an eigenvalue; all nonzero multiples of v are corresponding eigenvectors.

Exercise 5.1.3: Check a specified vector directly

The candidate is $v = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, which is nonzero. Multiplying by the source matrix gives

$$Av = \begin{bmatrix} 1 & -1 \\ 6 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 \\ -6 \end{bmatrix}.$$

This equals $-2v$ in every coordinate. Therefore v is an eigenvector with eigenvalue $\boxed{-2}$.

Exercise 5.1.4: Check a specified vector directly

The candidate is $v = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, which is nonzero. Multiplying by the source matrix gives

$$Av = \begin{bmatrix} 5 & 2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 3 \end{bmatrix}.$$

This equals $3v$ in every coordinate. Therefore v is an eigenvector with eigenvalue $\boxed{3}$.

Exercise 5.1.5: Check a specified vector directly

The candidate is $v = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$, which is nonzero. Multiplying by the source matrix gives

$$Av = \begin{bmatrix} -4 & 3 & 3 \\ 2 & -3 & -2 \\ -1 & 0 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -15 \\ 10 \\ -5 \end{bmatrix}.$$

This equals $-5v$ in every coordinate. Therefore v is an eigenvector with eigenvalue $\boxed{-5}$.

Exercise 5.1.6: Check a specified vector directly

The candidate is $v = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}$, which is nonzero. Multiplying by the source matrix gives

$$Av = \begin{bmatrix} 3 & 6 & 7 \\ 3 & 2 & 7 \\ 5 & 6 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 13 \\ 1 \end{bmatrix}.$$

The first coordinate would force $\lambda = 5$, but the second would then have to be -10, not 13. A single common multiplier must work in every coordinate, so v is not an eigenvector.

Exercise 5.1.7: Produce a nonzero vector in the shifted null space

Solve $(A - 4I)x = 0$; the coefficient reduction is

$$A - 4I = \begin{bmatrix} -1 & 0 & -1 \\ 2 & -1 & 1 \\ -3 & 4 & 1 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_2$, $\quad R_1 \leftarrow \frac{1}{2}R_1$, $\quad R_2 \leftarrow R_2 - (-1)R_1$, $\quad R_3 \leftarrow R_3 - (-3)R_1$. Then

$$\begin{bmatrix} 1 & -\frac{1}{2} & \frac{1}{2} \\ 0 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{5}{2} & \frac{5}{2} \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3$, $R_2 \leftarrow \frac{2}{5}R_2$, $R_1 \leftarrow R_1 - (-\frac{1}{2})R_2$, $R_3 \leftarrow R_3 - (-\frac{1}{2})R_2$. Then

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The free variable gives, for example, $v = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$. This is nonzero, and substitution gives $Av = \begin{bmatrix} -4 \\ -4 \\ 4 \end{bmatrix} = 4v$.

Conclusion. Yes. The specified scalar is an eigenvalue and v is a corresponding eigenvector.

Exercise 5.1.8: Produce a nonzero vector in the shifted null space

Solve $(A - 1I)x = 0$; the coefficient reduction is

$$A - 1I = \begin{bmatrix} 3 & -2 & 3 \\ 0 & -2 & 3 \\ -1 & 2 & -3 \end{bmatrix}.$$

Use $R_1 \leftarrow \frac{1}{3}R_1$, $R_3 \leftarrow R_3 - (-1)R_1$. Then

$$\begin{bmatrix} 1 & -\frac{2}{3} & 1 \\ 0 & -2 & 3 \\ 0 & \frac{4}{3} & -2 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3$, $R_2 \leftarrow \frac{3}{4}R_2$, $R_1 \leftarrow R_1 - (-\frac{2}{3})R_2$, $R_3 \leftarrow R_3 - (-2)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\frac{3}{2} \\ 0 & 0 & 0 \end{bmatrix}.$$

The free variable gives, for example, $v = \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix}$. This is nonzero, and substitution gives $Av = \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix} = 1v$.

Conclusion. Yes. The specified scalar is an eigenvalue and v is a corresponding eigenvector.

Exercise 5.1.9: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = 1$.

$$A - 1I = \begin{bmatrix} 2 & 0 \\ 2 & 0 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_1 = \text{Span}\left\{\begin{bmatrix}0\\1\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 1v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

The value $\lambda = 3$.

$$A - 3I = \begin{bmatrix}0 & 0\\2 & -2\end{bmatrix}, \qquad \text{rref}(A - \lambda I) = \begin{bmatrix}1 & -1\\0 & 0\end{bmatrix}.$$

Reading the free variables gives

$$E_3 = \text{Span}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 3v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

Exercise 5.1.10: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix}-4 & 2\\3 & 1\end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = -5$.

$$A - (-5)I = \begin{bmatrix}1 & 2\\3 & 6\end{bmatrix}, \qquad \text{rref}(A - \lambda I) = \begin{bmatrix}1 & 2\\0 & 0\end{bmatrix}.$$

Reading the free variables gives

$$E_{-5} = \text{Span}\left\{\begin{bmatrix}-2\\1\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = -5v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

Exercise 5.1.11: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix}1 & -3\\-4 & 5\end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = -1$.

$$A - (-1)I = \begin{bmatrix}2 & -3\\-4 & 6\end{bmatrix}, \qquad \text{rref}(A - \lambda I) = \begin{bmatrix}1 & -\frac{3}{2}\\0 & 0\end{bmatrix}.$$

Reading the free variables gives

$$E_{-1} = \text{Span}\left\{\begin{bmatrix}3\\2\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = -1v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

The value $\lambda = 7$.

$$A - 7I = \begin{bmatrix}-6 & -3\\-4 & -2\end{bmatrix}, \qquad \text{rref}(A - \lambda I) = \begin{bmatrix}1 & \frac{1}{2}\\0 & 0\end{bmatrix}.$$

Reading the free variables gives

$$E_{7} = \text{Span}\left\{\begin{bmatrix}-1\\2\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 7v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

Exercise 5.1.12: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix}4 & 1\\3 & 6\end{bmatrix}$, solve $(A-\lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = 3$.

$$A - 3I = \begin{bmatrix}1 & 1\\3 & 3\end{bmatrix}, \qquad \text{rref}(A - \lambda I) = \begin{bmatrix}1 & 1\\0 & 0\end{bmatrix}.$$

Reading the free variables gives

$$E_{3} = \text{Span}\left\{\begin{bmatrix}-1\\1\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 3v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

The value $\lambda = 7$.

$$A - 7I = \begin{bmatrix}-3 & 1\\3 & -1\end{bmatrix}, \qquad \text{rref}(A - \lambda I) = \begin{bmatrix}1 & -\frac{1}{3}\\0 & 0\end{bmatrix}.$$

Reading the free variables gives

$$E_{7} = \text{Span}\left\{\begin{bmatrix}1\\3\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 7v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

Exercise 5.1.13: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = 1$.

$$A - 1I = \begin{bmatrix} 3 & 0 & 1 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_1 = \operatorname{Span}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 1v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

The value $\lambda = 2$.

$$A - 2I = \begin{bmatrix} 2 & 0 & 1 \\ -2 & -1 & 0 \\ -2 & 0 & -1 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_2 = \operatorname{Span}\left\{ \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} \right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 2v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

The value $\lambda = 3$.

$$A - 3I = \begin{bmatrix} 1 & 0 & 1 \\ -2 & -2 & 0 \\ -2 & 0 & -2 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 3v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

Exercise 5.1.14: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix} 4 & 0 & -1 \\ 3 & 0 & 3 \\ 2 & -2 & 5 \end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = 3$.

$$A - 3I = \begin{bmatrix} 1 & 0 & -1 \\ 3 & -3 & 3 \\ 2 & -2 & 2 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 3v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

Exercise 5.1.15: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix} -4 & 1 & 1 \\ 2 & -3 & 2 \\ 3 & 3 & -2 \end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = -5$.

$$A - (-5)I = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_{-5} = \operatorname{Span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent because they arise from distinct choices of a free coordinate; the pivot equations show that they span all solutions. Each satisfies $Av = -5v$ by direct substitution. Thus the displayed list is a basis, of dimension 2.

Exercise 5.1.16: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix} 5 & 0 & -1 & 0 \\ 1 & 3 & 0 & 0 \\ 2 & -1 & 3 & 0 \\ 4 & -2 & -2 & 4 \end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = 4$.

$$A - 4I = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 1 & -1 & 0 & 0 \\ 2 & -1 & -1 & 0 \\ 4 & -2 & -2 & 0 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_4 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent because they arise from distinct choices of a free coordinate; the pivot equations show that they span all solutions. Each satisfies $Av = 4v$ by direct substitution. Thus the displayed list is a basis, of dimension 2.

Exercise 5.1.17: Read a triangular matrix's eigenvalues

The source matrix is upper triangular with diagonal entries $0, 3, -2$. For an upper triangular matrix, the eigenvalues are precisely its diagonal entries: $A - \lambda I$ loses a pivot exactly when one of these entries minus λ becomes zero.

Conclusion. The eigenvalues are $0, 3, -2$.

Exercise 5.1.18: The lower-triangular case

The source matrix is lower triangular with diagonal entries $5, 0, 3$. The triangular-matrix theorem therefore gives the eigenvalues directly.

Conclusion. The eigenvalues are $5, 0, 3$. Off-diagonal entries do not change this conclusion for a triangular matrix.

Exercise 5.1.19: A singular matrix has eigenvalue zero

All three columns are multiples of $(1, 1, 1)^T$, so they are dependent. Thus $Ax = 0$ has a nonzero solution and 0 is an eigenvalue. For example $(-2, 1, 0)^T$ gives a zero column combination, without needing elimination. The problem asks for one eigenvalue, not the complete list.

Exercise 5.1.20: Two independent vectors with coordinate sum zero

Each column of A is $(2,2,2)^T$. Therefore any input whose coordinates sum to zero is sent to zero. In particular

$$v_1 = (1,-1,0)^T, \qquad v_2 = (1,0,-1)^T$$

are eigenvectors for $\lambda = 0$. They are independent: in $av_1 + bv_2 = 0$, coordinate two forces $a = 0$ and coordinate three forces $b = 0$.

Conclusion. One eigenvalue is 0, with the two independent eigenvectors above.

Exercise 5.1.21: Do not omit the nonzero-vector requirement

(a) **False.** Taking $x = 0$ makes $Ax = \lambda x$ true for any scalar, even one that is not an eigenvalue. The definition requires $x \neq 0$.

(b) **True.** Noninvertibility is equivalent to $Ax = 0$ having a nonzero solution, precisely an eigenvector for eigenvalue zero.

(c) **True.** Rearranging $Ax = cx$ gives $(A - cI)x = 0$, with the same nonzero x required in both formulations.

(d) **True.** Compute Ax and check whether it equals one scalar multiple of the specified nonzero x. Finding the vector from scratch is a different task.

(e) **False.** Row operations on A need not preserve its eigenvalues. For example $\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$ has eigenvalues $2, 0$, but replacing row two by row two minus row one gives triangular $\begin{bmatrix}1 & 1\\ 0 & 0\end{bmatrix}$, with eigenvalues $1, 0$.

Exercise 5.1.22: Eigenspaces and repeated eigenvalues

(a) **False.** Zero solves the equality but is not an eigenvector. The condition $x \neq 0$ is missing.

(b) **False.** The independent vectors e_1, e_2 are both eigenvectors of I_2 with the same eigenvalue 1.

(c) **True.** A steady-state probability vector q is nonzero and satisfies $Mq = q$, so it is an eigenvector for eigenvalue 1.

(d) **False in general.** The matrix $\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$ has eigenvalues $1, -1$, not the diagonal entry 0. The diagonal rule applies to triangular matrices.

(e) **True.** $E_\lambda = \mathrm{Nul}(A - \lambda I)$ by rearranging the eigenvector equation; the null space also includes zero.

Exercise 5.1.23: Bound the number of distinct eigenvalues without a polynomial

If an $n \times n$ matrix had p distinct eigenvalues, choose one nonzero eigenvector for each. Theorem 2 says those p vectors are independent. A list of vectors in $\mathbb{R}^n$ cannot be independent when $p > n$.

Hence $p \leq n$. In particular a 2×2 matrix has at most two distinct eigenvalues. This bounds distinct values, not the number of different eigenvectors, which can be infinite.

Exercise 5.1.24: An example with one distinct eigenvalue

Take $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2I_2$. Every nonzero vector satisfies $Ax = 2x$. If $Ax = \lambda x$ with $x \neq 0$, then $(2 - \lambda)x = 0$ forces $\lambda = 2$. Thus the only distinct eigenvalue is 2, although there are infinitely many corresponding eigenvectors.

Exercise 5.1.25: Invert the eigenvalue as well as the matrix

Choose $x \neq 0$ with $Ax = \lambda x$. Since A is invertible, $\lambda \neq 0$: otherwise $Ax = 0$ would force $x = 0$. Multiply the eigenvector equation by A^{-1}: $x = \lambda A^{-1}x$, so $A^{-1}x = \lambda^{-1}x$ after division by the nonzero scalar. The vector x is still nonzero, proving that λ^{-1} is an eigenvalue of A^{-1} with the same eigenvector.

Exercise 5.1.26: A square-zero matrix can have only eigenvalue zero

If $Ax = \lambda x$ with $x \neq 0$, applying A again gives $0 = A^2x = \lambda^2 x$. A nonzero vector times a scalar is zero only when the scalar is zero, so $\lambda = 0$. Zero really is an eigenvalue: if $A = 0$, every nonzero vector works; if $A \neq 0$, choose y with $Ay \neq 0$ and set $x = Ay$. Then $x \neq 0$ but $Ax = A^2y = 0$.

Conclusion. The only eigenvalue is 0.

Exercise 5.1.27: Transpose preserves eigenvalues

The identity $A^T - \lambda I = (A - \lambda I)^T$ is the key. A matrix B is invertible if and only if B^T is invertible: if B^{-1} exists, $(B^{-1})^T$ is an inverse of B^T, and applying the same argument to B^T proves the converse. Therefore $A - \lambda I$ is singular exactly when $A^T - \lambda I$ is singular. Singularity of the shifted matrix is equivalent to the existence of a nonzero eigenvector.

Conclusion. A and A^T have the same eigenvalues, though their eigenvectors need not agree.

Exercise 5.1.28: Deduce the lower-triangular theorem by transposing

If A is lower triangular, A^T is upper triangular and has the same diagonal entries. The established upper-triangular case says that the eigenvalues of A^T are precisely these entries. Exercise 27 says A has exactly the same eigenvalues, completing the lower-triangular case without repeating an elimination argument.

Exercise 5.1.29: Equal row sums produce an obvious eigenvector

Let $\mathbf{1} = (1, \ldots, 1)^T \neq 0$. Entry i of $A\mathbf{1}$ is the sum of row i, which is s by hypothesis. Thus $A\mathbf{1} = (s, \ldots, s)^T = s\mathbf{1}$.

Conclusion. s is an eigenvalue, with eigenvector $\mathbf{1}$.

Exercise 5.1.30: Equal column sums use the transpose

The column sums of A are the row sums of A^T. Exercise 29 therefore makes s an eigenvalue of A^T. By Exercise 27 it is also an eigenvalue of A. The all-ones vector is an eigenvector of A^T here; it need not itself be an eigenvector of A, so the transpose step is substantive.

Exercise 5.1.31: Read eigenvalues from a reflection

Every nonzero vector on the reflecting line L is fixed, so $\lambda = 1$ is an eigenvalue and $E_1 = L$. A vector perpendicular to L is reversed, so -1 is the other eigenvalue, with $E_{-1} = L^{\perp}$. No vector outside L can be fixed: decompose it into its along-line and perpendicular parts; the reflection reverses the latter, so equality requires that part to be zero.

Exercise 5.1.32: Read a fixed eigenspace from a rotation

Every vector on the rotation axis L is fixed, so 1 is an eigenvalue. For a nontrivial rotation, the perpendicular-plane part of a nonzero vector is rotated and cannot remain unchanged, giving $E_1 = L$. If the angle is a whole multiple of 2π, the transformation is instead the identity and $E_1 = \mathbb{R}^3$. This qualification handles the degenerate zero-angle rotation, which the wording does not exclude.

Exercise 5.1.33: Combine two eigenvector solutions of a recurrence

(a) Shift the index in the definition.

The next term is $x_{k+1} = c_1\lambda^{k+1}u + c_2\mu^{k+1}v$.

(b) Apply A to the stated expression.

Using $Au = \lambda u$, $Av = \mu v$, and linearity,

$$Ax_k = c_1\lambda^k Au + c_2\mu^k Av = c_1\lambda^{k+1}u + c_2\mu^{k+1}v = x_{k+1}.$$

Thus the sequence satisfies the difference equation at every nonnegative index. Zeroth powers are taken as 1, including for a zero eigenvalue, as required by the initial term.

Exercise 5.1.34: Try expanding the initial state in eigenvectors

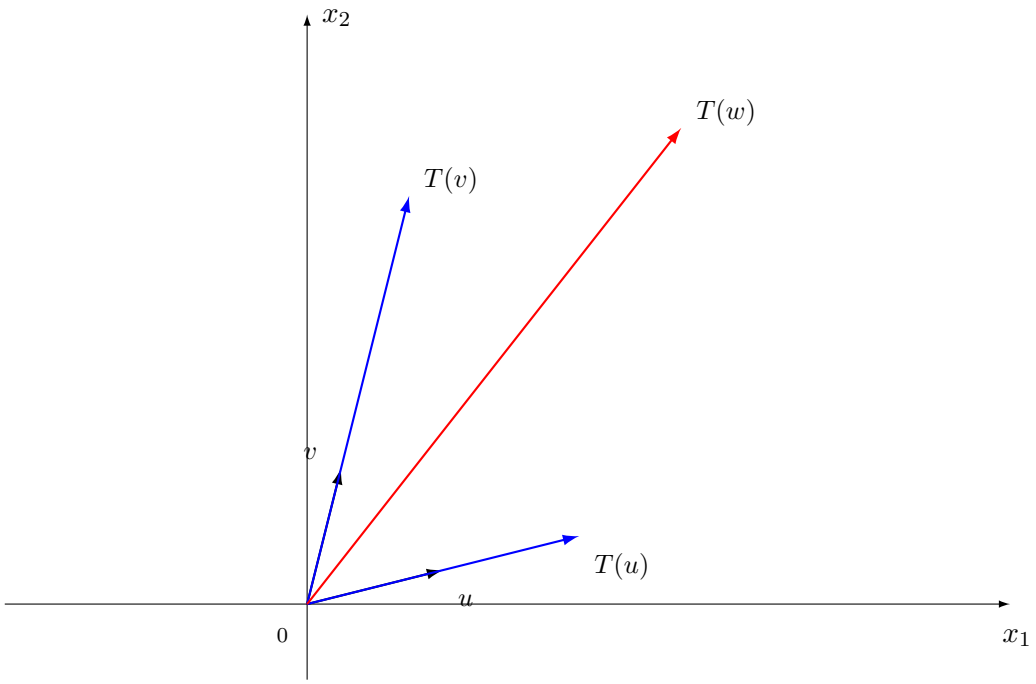
Find eigenvectors v_j of A and try solving $x_0 = \sum_j c_j v_j$ for coefficients c_j. If this is possible, set $x_k = \sum_j c_j \lambda_j^k v_j$, using the corresponding eigenvalues. At $k = 0$ it equals the prescribed state, and the calculation in Exercise 33 proves $Ax_k = x_{k+1}$ term by term. This method works whenever x_0 lies in the span of the available eigenvectors; in particular it works for every x_0 when there is an eigenvector basis. Not every matrix has such a basis, so this is a useful construction rather than an unconditional guarantee.

Exercise 5.1.35: Sketch the action on two eigendirections

The eigenvector equations give $T(u) = 2u$ and $T(v) = 3v$. Since $w = u + v$, linearity gives $T(w) = 2u + 3v$. Thus double the arrow along u, triple the arrow along v, and add those two new

arrows head to tail.

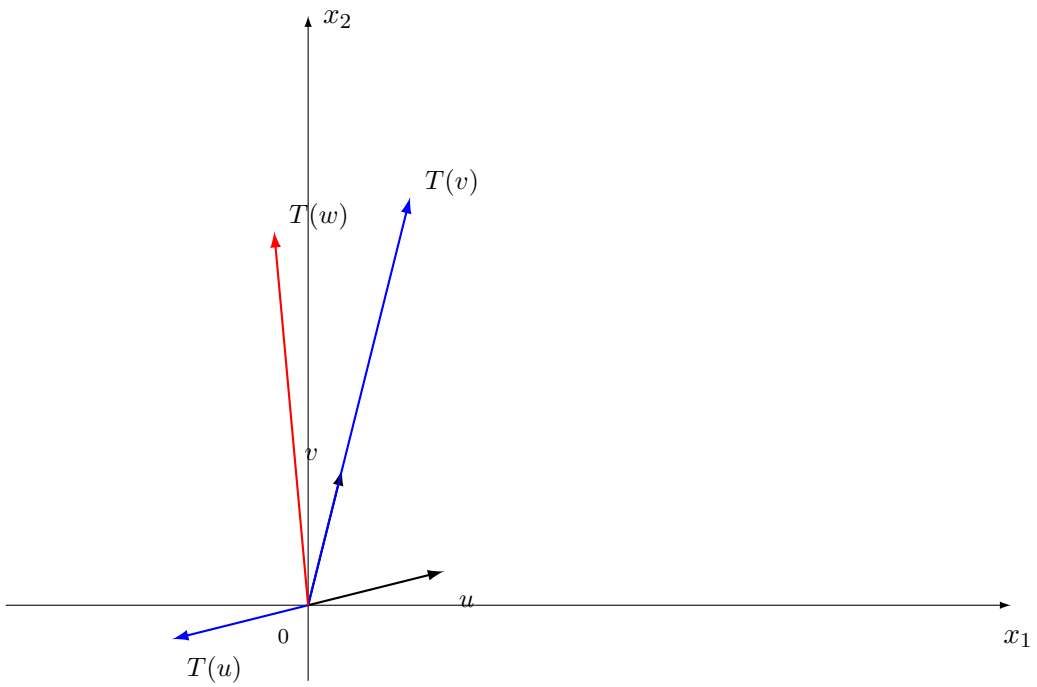
The black arrows reproduce the two source directions schematically; blue shows the separate images and red their sum. No numerical coordinates are assumed for the textbook's unlabeled vectors.



Exercise 5.1.36: Sketch the action on two eigendirections

The eigenvector equations give $T(u) = -1u$ and $T(v) = 3v$. Since $w = u + v$, linearity gives $T(w) = -1u + 3v$. Reverse the arrow u without changing its length, triple the arrow v, and add the resulting arrows head to tail.

The black arrows reproduce the two source directions schematically; blue shows the separate images and red their sum. No numerical coordinates are assumed for the textbook's unlabeled vectors.



Exercise 5.1.37: Compute eigenvalues, then reduce each shifted matrix

For the supplied matrix

$$A = \begin{bmatrix} 12 & 1 & 4 \\ 2 & 11 & 4 \\ 1 & 3 & 7 \end{bmatrix},$$

exact computation gives the factored characteristic polynomial

$$\det(A - \lambda I) = -(\lambda - 15)(\lambda - 10)(\lambda - 5).$$

Eigenvalue 5**.**

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_5 = \left\{ \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} \right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av = 5v$ by exact substitution. The eigenspace dimension is 1.

Eigenvalue 10**.**

$$\operatorname{rref}(A-(10)I)=\begin{bmatrix}1&0&3\\0&1&-2\\0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{10}=\left\{\begin{bmatrix}-3\\2\\1\end{bmatrix}\right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av=10v$ by exact substitution. The eigenspace dimension is 1.

Eigenvalue 15**.**

$$\operatorname{rref}(A-(15)I)=\begin{bmatrix}1&0&-2\\0&1&-2\\0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{15}=\left\{\begin{bmatrix}2\\2\\1\end{bmatrix}\right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av=15v$ by exact substitution. The eigenspace dimension is 1.

The listed roots exhaust the characteristic polynomial, so no eigenvalues are omitted. A repeated root need not supply the same number of independent eigenvectors as its multiplicity; the null-space calculation, not multiplicity alone, determines the basis.

Exercise 5.1.38: Compute eigenvalues, then reduce each shifted matrix

For the supplied matrix

$$A=\begin{bmatrix}5&-2&2&-4\\7&-4&2&-4\\4&-4&2&0\\3&-1&1&-3\end{bmatrix},$$

exact computation gives the factored characteristic polynomial

$$\det(A-\lambda I)=(\lambda-2)\,(\lambda-1)\,(\lambda+1)\,(\lambda+2)\,.$$

Eigenvalue -2**.**

$$\operatorname{rref}(A-(-2)I)=\begin{bmatrix}1&0&0&0\\0&1&-1&0\\0&0&0&1\\0&0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{-2}=\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}\right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av=-2v$ by exact substitution. The eigenspace dimension is 1.

Eigenvalue -1**.**

$$\operatorname{rref}(A-(-1)I)=\begin{bmatrix}1&0&0&-1\\0&1&0&-1\\0&0&1&0\\0&0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{-1}=\left\{\begin{bmatrix}1\\1\\0\\1\end{bmatrix}\right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av=-1v$ by exact substitution. The eigenspace dimension is 1.

Eigenvalue 1**.**

$$\operatorname{rref}(A-(1)I)=\begin{bmatrix}1&0&0&-2\\0&1&0&-2\\0&0&1&0\\0&0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{1}=\left\{\begin{bmatrix}2\\2\\0\\1\end{bmatrix}\right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av=1v$ by exact substitution. The eigenspace dimension is 1.

Eigenvalue 2**.**

$$\operatorname{rref}(A-(2)I)=\begin{bmatrix}1&0&0&-2\\0&1&0&-2\\0&0&1&-1\\0&0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_2=\left\{\begin{bmatrix}2\\2\\1\\1\end{bmatrix}\right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av=2v$ by exact substitution. The eigenspace dimension is 1.

The listed roots exhaust the characteristic polynomial, so no eigenvalues are omitted. A repeated root need not supply the same number of independent eigenvectors as its multiplicity; the null-space calculation, not multiplicity alone, determines the basis.

Exercise 5.1.39: Compute eigenvalues, then reduce each shifted matrix

For the supplied matrix

$$A=\begin{bmatrix}12&-90&30&30&30\\8&-49&15&15&15\\16&-52&12&0&20\\0&-30&10&22&10\\8&-41&15&15&7\end{bmatrix},$$

exact computation gives the factored characteristic polynomial

$$\det(A-\lambda I)=-(\lambda-12)^2(\lambda+4)(\lambda+8)^2.$$

Eigenvalue -8**.**

$$\operatorname{rref}(A-(-8)I)=\begin{bmatrix}1&0&0&-3&0\\0&1&0&-\frac{3}{2}&0\\0&0&1&-\frac{3}{2}&1\\0&0&0&0&0\\0&0&0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{-8}=\left\{\begin{bmatrix}6\\3\\3\\2\\0\end{bmatrix},\ \begin{bmatrix}0\\0\\-1\\0\\1\end{bmatrix}\right\}.$$

Its 2 vectors are independent, span every solution of the reduced system, and satisfy the original $Av = -8v$ by exact substitution. The eigenspace dimension is 2.

Eigenvalue -4**.**

$$\operatorname{rref}(A - (-4)I) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{-4} = \left\{ \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av = -4v$ by exact substitution. The eigenspace dimension is 1.

Eigenvalue 12**.**

$$\operatorname{rref}(A - (12)I) = \begin{bmatrix} 1 & 0 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{12} = \left\{ \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Its 2 vectors are independent, span every solution of the reduced system, and satisfy the original $Av = 12v$ by exact substitution. The eigenspace dimension is 2.

The listed roots exhaust the characteristic polynomial, so no eigenvalues are omitted. A repeated root need not supply the same number of independent eigenvectors as its multiplicity; the null-space calculation, not multiplicity alone, determines the basis.

Exercise 5.1.40: Compute eigenvalues, then reduce each shifted matrix

For the supplied matrix

$$A = \begin{bmatrix} -23 & 57 & -9 & -15 & -59 \\ -10 & 12 & -10 & 2 & -22 \\ 11 & 5 & -3 & -19 & -15 \\ -27 & 31 & -27 & 25 & -37 \\ -5 & -15 & -5 & 1 & 31 \end{bmatrix},$$

exact computation gives the factored characteristic polynomial

$$\det(A - \lambda I) = -(\lambda - 42)^2 (\lambda + 14)^3.$$

Eigenvalue -14**.**

$$\operatorname{rref}(A - (-14)I) = \begin{bmatrix} 1 & 0 & 1 & 0 & -\frac{10}{3} \\ 0 & 1 & 0 & 0 & -2 \\ 0 & 0 & 0 & 1 & -\frac{5}{3} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{-14} = \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 10 \\ 6 \\ 0 \\ 5 \\ 3 \end{bmatrix} \right\}.$$

Its 2 vectors are independent, span every solution of the reduced system, and satisfy the original $Av = -14v$ by exact substitution. The eigenspace dimension is 2.

Eigenvalue 42**.**

$$\operatorname{rref}(A - (42)I) = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & -\frac{1}{5} & \frac{1}{5} \\ 0 & 0 & 1 & \frac{2}{5} & \frac{3}{5} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{42} = \left\{ \begin{bmatrix} 0 \\ 1 \\ -2 \\ 5 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ -1 \\ -3 \\ 0 \\ 5 \end{bmatrix} \right\}.$$

Its 2 vectors are independent, span every solution of the reduced system, and satisfy the original $Av = 42v$ by exact substitution. The eigenspace dimension is 2.

The listed roots exhaust the characteristic polynomial, so no eigenvalues are omitted. A repeated root need not supply the same number of independent eigenvectors as its multiplicity; the null-space calculation, not multiplicity alone, determines the basis.